



MIT SLOAN SCHOOL OF MANAGEMENT
MIT SCHWARZMAN COLLEGE OF COMPUTING



MODULE 1 UNIT 1
Lesson Video 1 Transcript

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The AI continuum

PAUL MCDONAGH-SMITH: The story of AI is often told as a single narrative, but it's more useful to think of it as a continuum, a series of expanding capabilities, each one building on what came before.

It begins with classical AI and machine learning, systems that learn patterns from data, a recommendation engine that suggests what you might buy next, a fraud detection model that flags unusual transactions. These systems are powerful, but they operate within tightly defined boundaries. They classify, they predict, they respond.

Then came deep learning, neural networks capable of processing far more complex inputs, from images to speech to text. And then by the early twenty twenties, we arrived at generative AI. Large language models like GPT and Claude and Gemini that can produce remarkably fluent text, images and code. These models represented a genuine inflection point. But here's the critical distinction, and I want you to hold on to this. Generative AI at its core is still reactive. You prompt it, it responds. It doesn't set its own goals, it doesn't plan a sequence of steps, it doesn't decide which tools to reach out for or when to ask for help.

What makes AI

PAUL MCDONAGH-SMITH: Agentic AI is the next threshold on that continuum, and it's a qualitative shift, not merely a quantitative one. Think of it this way. If a standard large language model is like a research assistant who answers whatever you ask, question in, question out, then an agentic AI system is more like a super capable project manager. You give it an objective, and it figures out the steps. It decomposes the task. It selects the right tools, it executes, checks its work, adjusts course, and when it hits a wall, it escalates to a human. That's the shift from systems that respond to systems that pursue goals. From reactive to proactive systems.

Agentic AI systems typically exhibit a core set of characteristics. They can reason about multi step problems. They can plan and decompose tasks into sequences of actions. They can use external tools, databases, APIs, search engines. They have a form of memory allowing them to maintain context across a workflow rather than treating every interaction as a blank slate. And they have feedback loops, the ability to evaluate their own outputs and refine them.

Now a word of calibration. Where we are today isn't artificial general intelligence. These systems still hallucinate. They still propagate errors across steps. They still require human oversight at critical junctures. The frontier is real, but so are the constraints, and understanding both is essential for making sound investment and adoption decisions. So where does the value sit?

Value and constraints

PAUL MCDONAGH-SMITH: Agentic AI is most compelling where workflows are multi-step, cross functional and involve judgement across a chain of decisions. Think of claims processing in insurance, where an agent could ingest a claim, verify policy details against a database, cross reference damage assessments, flag anomalies and draft a resolution, all without a human manually shepherding each step.

Or perhaps consider supply chain monitoring, where an agentic system could detect a disruption, evaluate alternatives, and trigger a procurement workflow continuously, not just when someone remembers to check a dashboard. The value isn't in replacing human judgment, it's in amplifying it. Handling the high-volume cognitive work so that human attention can be reserved for the decisions that truly require it.

But the constraints are equally important to name. Today's agentic systems remain brittle in unstructured environments. They perform best when the problem space is well defined, where the tools are well integrated, and where the guardrails are explicit. Autonomy without structure is risk.

Here's the takeaway I'd like you to carry forward.

Conclusion

PAUL MCDONAGH-SMITH: Agentic AI is an architectural capability you design. One that sits at a specific point on the AI continuum between the reactive systems most organizations already use and the fully autonomous systems that remain in large part still aspirational. Your job as a leader is to understand where on that continuum your organization's real opportunities live, and to build accordingly.

SPEAKER: Did you understand all the concepts covered in this video? If you'd like to review any of the content, please click on the chapter menu.